

Hoang-Loc Cao

📍 Ho Chi Minh, Vietnam | ✉ loc11052004@gmail.com | 🎓 [Google Scholar](#) | 🏠 [Homepage](#) | [LinkedIn](#) | [Github](#)

RESEARCH INTERESTS

My research focuses on building trustworthy AI systems that can operate reliably under real-world clinical and high-stakes constraints. I work at the intersection of **AI in healthcare** and **trustworthy agentic AI**, with two main research directions:

- **AI in healthcare:** robust and explainable learning from limited, noisy, and heterogeneous clinical data. My interests include affective and mental health computing from multimodal signals such as text, speech, visual, physiological, and neuroimaging data, as well as label-efficient medical image analysis under domain shift, imperfect and limited supervision.
- **Trustworthy agentic AI:** self-improving and contestable multi-agent systems for high-stakes decision-making, with an emphasis on neuro-symbolic reasoning, human-in-the-loop feedback, continual learning, and reinforcement learning.

EDUCATION

University of Science, VNU-HCM

Ho Chi Minh, Vietnam

Bachelor of Science | Major: Data Science

Sep 2022 – Sep 2026

- Thesis: Explainable AI for Text-Based Depression Detection (10.0/10.0).
- Advisor: [Prof. Bac Le](#).
- GPA: 3.63/4.0.

PROFESSIONAL EXPERIENCE

Machine and Hybrid Intelligence Lab, Northwestern University

Illinois, USA (Remote)

Research Fellow

Sep 2025 – Present

- Advisor: [Prof. Ulas Bagci](#).
- Research topics: medical image analysis, label-efficient learning.
- Developed SDT-Net, a dual-teacher framework for scribble-supervised medical image segmentation that addresses sparse annotation ambiguity via dynamic teacher switching and hierarchical feature consistency.
- Developed a decoupled training strategy addressing domain shift and class imbalance for White Blood Cell classification, achieving **3rd place** at WBCBench – IEEE ISBI 2026 Challenge.

Analytics Everywhere Lab, University of New Brunswick

Fredericton, Canada (Remote)

Research Assistant

Aug 2025 – Present

- Advisor: [Prof. Hung Cao](#).
- Research topics: explainable AI, agentic AI, AI for healthcare.
- Led the lab team in the [BIC-MAC Challenge](#) (MICCAI 2026) on multimodal 3D pseudo-CT synthesis for PET attenuation correction, achieving **1st place**.
- Designed self-evolving human-centered framework for explainable depression symptom annotation, reducing expert annotation overhead by **up to 75%** while improving detection of subtle clinical signals in social media text.
- Conducted research on building neuro-symbolic reasoning framework for multi-agent collaboration, applied for multiple domains including care planning, legal reasoning and multimedia verification.

Faculty of Information Technology, University of Science, VNU-HCM

Ho Chi Minh, Vietnam

Undergraduate Research Assistant

Sep 2025 – Present

- Advisor: [Prof. Bac Le](#).
- Conducted research on explainable AI for NLP, multi-agent systems, and self-evolving systems. Developed a computational argumentation-based multi-agent collaboration framework for explainable depression detection.

PASSIO Lab, North Carolina A&T State University

Remote

Research Assistant

Sep 2025 – Dec 2025

- Advisor: [Prof. Phat Kim Huynh](#).
- Assisted in a semi-supervised multi-modal brain tumor segmentation project by supporting the development and evaluation of the proposed method for MRI segmentation under limited labeled data.

Quy Nhon AI Center, FPT Software

Gia Lai, Vietnam

AI Engineer Intern

May 2025 – Aug 2025

- Implemented a trustworthy multilingual interview training system powered by Large Language Models.

PUBLICATIONS

C: Conference, J: Journal, P: Preprint/Submitted | * indicates equal contribution

- [C.1] T. H. Nguyen*, **Hoang-Loc Cao***, D. T. Chung, M. A. Vu, T. M. Nguyen, M. Le, P. K. Huynh, and U. Bagci (2026). **Scribble-Supervised Medical Image Segmentation with Dynamic Teacher Switching and Hierarchical Consistency**. *IEEE International Symposium on Biomedical Imaging (ISBI 2026)*. [DOI] [arXiv]
- [C.2] L. Le*, **Hoang-Loc Cao***, H. H. Pham, T. H. Nguyen, and U. Bagci (2026). **Robust White Blood Cell Classification with Stain-Normalized Decoupled Learning and Ensembling**. *IEEE International Symposium on Biomedical Imaging (ISBI 2026)*. [DOI] [arXiv]
- [C.3] **Hoang-Loc Cao***, V. Pham*, and B. Le (2026). **Argumentative Multi-Agent Debate for Explainable Depression Detection in Clinical Interviews**. *International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2026)*.
- [C.4] **Hoang-Loc Cao***, P. Ho*, T. T. H. Nguyen, P. T. L. Nguyen, D. T. L. Nguyen, and H. Cao (2026). **Neuro-Symbolic Adaptive Collaboration of Arena-Based Argumentative LLMs for Contestable Legal Reasoning**. *Canadian Conference on Artificial Intelligence (Canadian AI 2026)*. [DOI] [arXiv]
- [C.5] **Hoang-Loc Cao***, V. Pham*, T. T. H. Nguyen, P. T. L. Nguyen, P. Ho, V. Whitford, and H. Cao (2026). **Self-Evolving DSM-5 Annotation Framework for Explainable Major Depressive Disorder Prediction**. *IEEE International Conference on Omni-Layer Intelligent Systems (COINS 2026)*.
- [C.6] T. T. H. Nguyen*, V. T. K. Nguyen*, **Hoang-Loc Cao**, P. Ho, V. Pham, and H. Cao (2026). **Contestable Multi-Agent Debate with Arena-based Argumentative Computation for Multimedia Verification**. *ACM International Conference on Multimedia Retrieval (ICMR 2026)*. [DOI]
- [C.7] T. D. Chung*, B. T. Lam*, T. H. Nguyen, T. Nguyen, N. L. V. Vu, **Hoang-Loc Cao**, P. K. Huynh, and M. Xu (2026). **Modality-Specific Enhancement and Complementary Fusion for Semi-Supervised Multi-Modal Brain Tumor Segmentation**. *AAAI Bridge Program – AIMedHealth 2026*. [PDF]
- [C.8] T. T. H. Nguyen, **Hoang-Loc Cao**, P. Ho, P. T. L. Nguyen, R. Richard, and H. Cao (2026). **Adaptive Arena-based Contestable Argumentative Network-of-Experts for Open-Ended Care Plan Coordination**. *International Conference on Frontiers of Artificial Intelligence, Ethics, and Multidisciplinary Applications (FAIEMA 2026)*.
- [C.9] T. T. H. Nguyen, K. V. Q. Nguyen, **Hoang-Loc Cao**, T. Duong, P. Ho, V. Pham, L. Nguyen, and H. Cao (2026). **Consensus-based Agentic Large Language Model Framework for Harmonized Tariff Schedule Code Classification**. *International Conference Resilience by Technology and Design (RTD 2026)*.
- [C.10] T. T. H. Nguyen, T. D. Q. Nguyen, **Hoang-Loc Cao**, T. C. T. Tran, T. C. M. Truong, and H. Cao (2025). **SimInterview: Transforming Business Education through Large Language Model-Based Simulated Multilingual Interview Training System**. *International Conference on Economics, Finance, and Management (ICEFM 2025)*. [PDF]

HONORS AND AWARDS

- Sep 2026 – Rank 1st at BIC-MAC Challenge @ MICCAI 2026.
- Jun 2026 – Rank 1st at Multimedia Verification Challenge @ ACM ICMR 2026.
- Apr 2026 – Rank 3rd at WBCBench Challenge @ IEEE ISBI 2026. [CERT]

SKILLS

- **Programming Languages:** Python, C/C++, SQL
- **Frameworks:** PyTorch, Scikit-learn, Hugging Face, MONAI, OpenCV, Langchain, Langgraph
- **Tools:** Git, Linux, Conda, Jupyter Notebook, LaTeX, Cloud (AWS, Azure)
- **Languages:** Vietnamese (Native), English (Fluent; IELTS 7.0 in 2022).